Network Reconnaissance and Traffic Analysis Lab

(Using Nmap and Wireshark)

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1. Introduction

With the rapid growth of digital systems and network-based applications, ensuring network security has become a critical requirement. One of the primary steps in cybersecurity is identifying potential vulnerabilities within a system through network analysis.

This project focuses on performing network reconnaissance and traffic analysis using industry-standard tools such as Nmap and Wireshark. The aim is to understand how systems are exposed to threats and how data flows within a network.

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2. Objective

The objectives of this project are:

- To identify open ports and active services on a system
- To detect service versions and system details
- To capture and analyze real-time network traffic
- To understand communication protocols such as DNS, TCP, HTTP, and TLS
- To correlate scanning results with actual network activity

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3. Tools Used

Nmap

An open-source tool used for network discovery and security auditing. It helps in identifying open ports, running services, and system characteristics.

Wireshark

A powerful packet analyzer used to capture and inspect network traffic in real time, allowing detailed protocol-level analysis.

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4. Methodology

4.1 Network Information Gathering

The system's IP address was identified using the following command:

```
ipconfig
```

The IPv4 address obtained was used as the target for scanning. Connectivity was verified using:

```
ping <target-ip>
```

4.2 Nmap Scanning

Multiple scans were performed to gather detailed information:

1. Basic Scan

```
nmap <target-ip>
```

- Identifies open ports
- Detects active hosts

2. Service Version Detection

```
nmap -sV <target-ip>
```

- Detects services running on open ports
- Provides version information

3. Aggressive Scan

```
nmap -A <target-ip>
```

- Performs OS detection
- Runs scripts
- Provides detailed service insights

4.3 Wireshark Traffic Capture

1. Wireshark was opened and the active network interface (Wi-Fi/Ethernet) was selected
2. Packet capture was started
3. Network activity was generated by:
    - Browsing websites
    - Running ping commands
    - Performing Nmap scans
4. Capture was stopped after a few minutes
5. Filters were applied to analyze specific protocols

4.4 Protocol Filters Used

- **DNS (Domain Name System):** Resolves domain names to IP addresses
- **TCP (Transmission Control Protocol):** Establishes reliable communication
- **HTTP (HyperText Transfer Protocol):** Transfers unencrypted web data
- **TLS (Transport Layer Security):** Provides encrypted communication

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5. Scan Results

The Nmap scans revealed multiple open ports and active services on the system.

Open Ports Identified

Port	Service	Description
135	MSRPC	Windows Remote Procedure Call
445	SMB	File sharing service
1521	Oracle TNS	Oracle database listener
3306	MySQL	Database service
7070	Remote Access	AnyDesk/remote service

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6. Traffic Analysis Results

DNS Analysis

- Observed domain name resolution
- Queries and responses visible in plain text
- Reveals browsing activity

TCP Analysis

- Three-way handshake observed:
  - SYN → SYN-ACK → ACK
- Ensures reliable communication

HTTP Analysis

- Minimal HTTP packets observed
- Data transmitted in plain text
- Considered insecure

TLS Analysis

- Majority of traffic encrypted
- Payload not readable
- Ensures secure communication

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7. Integrated Analysis (Nmap + Wireshark Correlation)

This project combines network reconnaissance and traffic analysis to provide a complete understanding of system behavior.

The Nmap scans identified open ports and services such as SMB, MySQL, and remote access services. These represent potential entry points for attackers.

Wireshark analysis showed how these services communicate in real time through protocols such as DNS, TCP, HTTP, and TLS.

Correlation Findings

- Open ports discovered via Nmap correspond to active communication observed in Wireshark
- Services identified in scanning generate identifiable packet patterns
- Scanning activity can be detected in traffic as repeated connection attempts

Key Insight

- Nmap shows what is exposed
- Wireshark shows how it is used

Together, they provide a comprehensive view of network security.

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8. Risk Analysis

The presence of open ports increases the system's attack surface.

- Port 445 (SMB): Common target for ransomware attacks
- Database Ports (1521, 3306): Risk of unauthorized access
- Remote Access (7070): Potential unauthorized control
- Port 135 (RPC): May be exploited for remote execution

Unencrypted traffic (HTTP and DNS) also exposes sensitive information.

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9. Recommendations

- Close unused ports
- Use firewall rules to restrict access
- Replace HTTP with HTTPS
- Secure database services with authentication
- Regularly update systems and services
- Monitor network traffic for suspicious activity

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10. Conclusion

This project successfully demonstrated how network reconnaissance and traffic analysis can be combined to assess system security.

Using Nmap, open ports and services were identified, while Wireshark provided insights into real-time communication.

The integration of both tools offers a deeper understanding of vulnerabilities and network behavior, making it an essential approach in cybersecurity.

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11. Future Scope

- Perform vulnerability scanning using advanced scripts
- Integrate intrusion detection systems (IDS)
- Analyze encrypted traffic in depth
- Automate scanning and reporting